

SQL Interview Practice - 100 Questions

JOIN Questions (1-20)

- Find customers who never placed an order.
- Find employees who do not belong to any department.
- List departments with zero employees.
- Find employees earning more than their manager.
- Show customer name and total order amount.
- Find customers with more than 3 orders.
- Find employees and their manager department.
- Find orders placed by USA customers.
- Find department-wise highest salary.
- Find products never ordered.
- Find customers who ordered Electronics products.
- Find employees working in the same department.
- Find duplicate customer records by name.
- Find orders with missing customer records.
- Find all departments and average salary.
- Find products sold by category.
- Find top customer by revenue.
- Find customers whose last order was in 2023.
- Find employees whose manager is NULL.
- Find total sales by country.

GROUP BY / HAVING Questions (21-40)

- Count orders per customer.
- Find customers with total revenue above 5000.
- Find average salary by department.
- Find departments where average salary > 60000.
- Find total inventory value by category.
- Find categories with more than 10 products.
- Find countries with more than 5 customers.
- Find monthly sales total.
- Find year-wise sales total.
- Find maximum order amount per customer.
- Find minimum salary per department.
- Find departments with salary spend above 100000.

- Find products with average price above 50 by category.
- Count distinct customers per country.
- Find customers with average order value above 500.
- Find months with total sales above 10000.
- Find categories where total stock is below 50.
- Find employees count by manager.
- Find order count by order month.
- Find products having duplicate names.

Window Functions (41-60)

- Rank employees by salary.
- Rank employees by salary within department.
- Find top 2 salaries per department.
- Find second highest salary using DENSE_RANK.
- Find latest order per customer.
- Find first order per customer.
- Calculate running total of sales.
- Calculate running stock quantity by product ID.
- Use LAG to compare current order with previous order.
- Use LEAD to compare current salary with next salary.
- Find month-over-month sales growth.
- Find customers' first and last order amount.
- Find duplicate rows using ROW_NUMBER.
- Delete duplicates using ROW_NUMBER.
- Find top 3 products by price in each category.
- Calculate cumulative salary department-wise.
- Calculate moving average of sales for last 3 orders.
- Divide employees into 4 salary groups using NTILE.
- Find each order's percentage of total sales.
- Find employees earning above department average.

CTE / Subquery Questions (61-80)

- Find employees earning more than company average salary.
- Find employees earning more than department average salary.
- Find second highest salary using subquery.
- Find customers whose total order amount is above average.
- Find products priced above category average.
- Find departments with no employees using NOT EXISTS.

- Find customers who placed at least one order using EXISTS.
- Find customers who never ordered using NOT EXISTS.
- Find products never sold using subquery.
- Find latest order per customer using CTE.
- Find duplicate products using CTE.
- Find monthly revenue using CTE.
- Find months where revenue is above average monthly revenue.
- Find top customer by revenue using CTE.
- Find categories where average price > 50 using CTE.
- Find employees whose salary is greater than manager salary.
- Find customers who bought from all categories.
- Find products with price greater than overall average.
- Find departments where max salary is above company average.
- Find orders above average order value.

Date / String / Advanced (81-100)

- Find orders placed in last 7 days.
- Find orders placed in current month.
- Find orders placed in 2024.
- Extract year, month, and day from order date.
- Find customers whose name starts with A.
- Find customers whose name ends with n.
- Find customers whose name contains 'ar'.
- Convert customer names to uppercase.
- Replace NULL country with Unknown.
- Create salary class: High, Medium, Low.
- Create order category: High Value if amount > 1000.
- Find products where name is NULL.
- Use COALESCE for missing product names.
- Find orders between two dates.
- Find employees hired in the last 1 year.
- Write query for incremental load using updated_date.
- Find duplicate records based on business key.
- Find records in source table but missing in target table.
- Compare source and target row counts by date.
- Find rejected records where mandatory columns are NULL.